

Important Directions

- **Read through all slides**
 - **Watch all videos**

Physical Development in Middle Childhood

Physical Growth

- Growth slow and steady
- Boys slightly taller and more muscular
- Lowest body mass index during this time

Gross Motor Development

- Advancement occurs in balance, strength, coordination, agility, and reaction time
- Involvement in organized sports increases

Fine motor skills

- Increased ability in fine motor skills
- Writing improves and becomes smaller and neater
- Fine motor skills will reach adult maturity by end of middle childhood
- Gross motor skills will continue to develop

Health issues

- Nutrition problems
- Overweight: BMI over 18
 - Obesity: BMI over 21

Obesity

- In the past 30 years, the occurrence of overweight in children has tripled.
- One in five children in the US is overweight (17%). (Fryar et al., 2012)
 - The most common prevalent nutritional disorder of US children and adolescents
 - One of the most common problems seen by pediatricians.

Obesity

- What is this epidemic?
- http://www.ted.com/talks/jamie_oliver

What other factors may play a role?

- **Sedentary lifestyle-**
 - Children who watched a lot of T.V grew up to be overweight and obese adults
 - Longitudinal study of children between 4-11 years of age in U.S
 - Those who watched 3 hours of TV a day gained 40% more body weight over time than those who watched less than 1.5 hours a day (Rideout, 2013)
 - This effect is stronger among ethnicities herein children tend to watch most TV per day (Rideout, 2013)
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- **What about fetal learning?**
 - Maternal malnutrition during pregnancy-



Genetics and Obesity

- Risk is inherited. But genes are not destiny
- The environment is crucial for determining outcome
- Epigenetics studies point to this.
- Naturalistic study of Pima Indians in Arizona and Mexico (Gladwell, 1998)
- Pima Indians of Mexico maintained their traditional diet rich in vegetables and low in sugars and fats
- Pima Indians of Arizona have changed their diets- mainstream US diet
- Pima Indian of AZ BMI is on avg. 50% higher than that of Pima Indians of Mexico

Consequences

- Link between obesity and low self- esteem and depression. However, direction of influence is not clear.
- Greater effect among girls than boys.
- Teasing disrupts academic performance.
- One study found obese girls lesser likely to enter college than their non-obese counterparts.
- Physical problems: Hypertension, diabetes, high cholesterol.

Prevention

- Multifaceted efforts involving parents, the mass media, food and beverage manufacturers, schools, restaurants and communities.

Hygiene Hypothesis: Asthma

- Rates of asthma are highest in middle childhood and increasing worldwide (Greenwood, 2011)
- Hygiene hypothesis explains this trend
- Children need healthy exposure to microbes to build immune systems
- Too much cleanliness is disadvantageous in this regard
- <https://www.youtube.com/watch?v=2fVcmn1Pdkc>

Cognitive Development in Middle Childhood

Piaget's theory

- Is thinking illogical in pre-operational stage?
- Why? What example comes to mind?
- Conservation tasks
- In concrete operations- Thinking is logical
- Children learn Conservation
- <https://www.youtube.com/watch?v=gA04ew6Oi9M>
- Only limited to concrete objects- Not abstract concepts
- <https://www.youtube.com/watch?v=3F1O4BGgW64>
- What is missing in the concrete operations stage?
- Ability to think in abstract terms. Logical thinking but tied to concrete objects.

INFORMATION-PROCESSING VIEW

- Question of Interest
 - What is it about the ways that middle school children think about things that make them better problem solvers than younger children?
 - Increases myelination in the brain increasing the speed of processing
 - Attention increments
 - Memory increments

INFORMATION-PROCESSING VIEW

- Selective attention-Focus on something while disregarding everything else
- Working memory and long-term memory
- Mnemonics-
 - ☐ Rehearsal strategies first emerge in middle childhood
 - ☐ Organization also used more often by school aged children than younger children
- Metamemory also improves-more realistic view of memory abilities

Theories of Intelligence

- Sternberg's Triarchic Theory
- Gardner's Theory of Multiple Intelligences

Componential (Analytic) Intelligence

- Information-processing mechanisms used in any context
 - ▣ Executive Functioning
 - ▣ Speed of processing and Working Memory
 - ▣ Knowledge

Experiential (Creative) Intelligence

- How people deal with novelty
 - Ability to generate new ideas and solve novel problems
- Ease with which we can automatize processing of recurrent stimuli or problems
 - Reading is a perfect example

Contextual (Practical) Intelligence

- Intelligence is not the same for a child in Chico as for one growing up in the slums of Rio De Janeiro, Brazil
 - Street smarts or **social intelligence**



"YOU CAN'T BUILD A HUT, YOU DON'T KNOW HOW TO FIND EDIBLE ROOTS AND YOU KNOW NOTHING ABOUT PREDICTING THE WEATHER. IN OTHER WORDS, YOU DO TERRIBLY ON OUR I.Q. TEST."

Contextual (Practical) Intelligence



Individual differences within each cultures can be evaluated by quality of

- **Adaptation**-adjusting behavior to fit environment
- **Selection**-selecting a different environment that's more suitable
- **Shaping**-changing the behavior and reactions of others to change the environment

Gardner's Theory of Multiple Intelligences

table 10.5 A List of Multiple Intelligences in Gardner's Theory: Their Definition, People Who Exemplify Them, and Their Theorized Neurological Locus

Intelligence	Definition	People Who Exemplify This Intelligence	Neurological System Hypothesized to Be Associated with This Intelligence
Linguistic	Sensitive to meaning and order of words	Maya Angelou (poet) Stephen King (writer) Martin Luther King (civil rights leader and orator)	Left hemisphere, temporal and frontal lobes
Logical-mathematical	Ability to reason logically and recognize patterns and order	Bill Gates (former CEO of Microsoft) Stephen Hawking (physicist) James Watson (biologist)	Left parietal lobe; left hemisphere for verbal naming; right hemisphere for spatial organization; frontal system for planning
Musical	Sensitivity to pitch, melody, rhythm, and tone	Yo Yo Ma (cellist) Mariah Carey (singer) Wolfgang Amadeus Mozart (composer)	Right anterior temporal lobe; frontal lobes
Bodily-kinesthetic	Ability to use one's body skillfully and handle objects adroitly	David Copperfield (magician) Kobe Bryant (basketball player) Mia Hamm (soccer player)	Cerebral motor strip; thalamus; basal ganglia; cerebellum
Spatial	Ability to perceive physical environment accurately and to recreate or transform aspects of that environment	Frank Lloyd Wright (architect) Pablo Picasso (painter) Georgia O'Keeffe (painter)	Right hemisphere, parietal occipital lobe
Naturalist	Ability to recognize and classify numerous species of flora and fauna	Charles Darwin (biologist) Jane Goodall (primatologist) E. O. Wilson (biologist)	Left parietal lobe (discriminating living from nonliving)
Interpersonal	Ability to understand people and relationships	Bill Clinton (politician) Ronald Reagan (politician) Madeline Albright (diplomat)	Frontal lobes
Intrapersonal	Access to one's emotional life as a means to understand oneself and others	Oprah Winfrey (talk show host) Bono (singer, philanthropist)	Frontal lobes
Spiritual/existential	Individuals who exhibit the propensity to pose (and ponder) questions about life, death, and ultimate realities	Albert Einstein (scientist) Socrates (philosopher) Dalai Lama (Tibetan monk)	Hypothesized as specific region in the right temporal lobe

School and Education across cultures: watch Revel video